Avi Sharma

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Education

Syracuse University, M.S. in Applied Data Science — May 2026

- **Coursework:** Applied Machine Learning, Natural Language Processing, Business Analytics, Advanced Database Management, Quantitative Analysis, Deep Learning, Human-Centered AI

Narsee Monjee Institute of Management Studies, B.Tech. in Computer Science — May 2024

- **Coursework:** Data Structures & Algorithms, Probability & Statistics, Machine Learning, Data Mining, Database Management Systems

Experience

Data Science Intern – MoMacMo — Sep 2025 – Present

- Built cloud pipelines on AWS (S3, EC2) to ingest and analyze 3.6+ TB of high-frequency sensor data end-to-end, from scratch with no existing infrastructure.
- Performed exploratory data analysis and statistical profiling on complex spatiotemporal datasets to surface patterns, anomalies, and data-quality issues that informed downstream modeling decisions.
- Designed inference workflows using VGG16 transfer learning and XGBoost for event detection and classification, achieving >90% accuracy matching expert benchmarks; translated raw signals into validated insights and visualizations for stakeholders.

AI Researcher – Healthcare AI – Syracuse University — Jan 2025 – Present

- Developed deep learning models for medical image classification using CNN architectures and transfer learning; ran structured error analysis to diagnose model weaknesses and iterate on data preprocessing and augmentation.
- Built reproducible training and evaluation pipelines with performance tracking, validation metrics, and A/B-style comparisons across model variants.
- Collaborated with cross-functional research teams and translated model outputs into clear insights for stakeholders; contributed to a project awarded **Best Project – Fall 2025**.

Projects

DataThon 26 – Syracuse Code Violations & Property Risk | *Python, SQL, GeoPandas, Scikit-learn, Folium* — Feb 2026

- Merged 140K+ code-violation records with 41K+ city assessment parcels on normalized SBL keys, building a clean parcel-level dataset for downstream analysis and modeling.
- Conducted EDA on violation volume, complaint types, neighborhoods, and open/closed status to surface temporal and spatial trends influencing property risk.
- Built a scikit-learn regression pipeline (ColumnTransformer + OneHotEncoder) to model total assessment value from violation counts and property context, exposing feature-level drivers of value.
- Partnered on Track B to construct a city grid layered with crime, violation, and vacancy signals; trained a Random Forest and shipped interactive Folium risk and vacancy-probability maps via GitHub Pages for public stakeholders.

MediExplain – Agentic AI Service | *LLMs, RAG, ChromaDB, Streamlit* — Aug 2025 – Dec 2025

- Designed the service architecture and inter-agent routing logic across an 18-component agentic system, iterating on user-facing response flows to improve interpretability and output reliability.
- Built the data layer using a ChromaDB vector store with a curated corpus, implementing semantic retrieval to ground responses in verified evidence and reduce hallucination.
- Engineered reliability layers (JSON schema validation, retry logic, output consistency checks) and defined evaluation metrics to monitor stability across varied and adversarial inputs.
- Delivered a production-facing Streamlit interface with privacy-safe data handling and user-adaptive response flows.

Northwind Data Warehouse & BI Analytics | *Snowflake, SQL, Power BI* — Oct 2025 – Dec 2025

- Built an end-to-end data warehouse in Snowflake with dimensional modeling (STG → DW) and SQL ETL pipelines to clean, transform, and integrate large datasets for downstream analysis.
- Analyzed customer purchasing behavior, order patterns, and operational KPIs to identify trends, inefficiencies, and growth opportunities.
- Designed interactive Power BI dashboards and defined KPIs for revenue, delivery, and inventory, translating complex data into actionable insights for stakeholders.

Driver Drowsiness Detection using Deep Learning | *Python, CNN, OpenCV, LSTM* — Nov 2025 – Dec 2025

- Built a real-time detection system using computer vision and deep learning to classify eye state and identify sustained drowsiness events from video.
- Trained and evaluated multiple architectures on 85K+ labeled images, selecting an augmented CNN achieving ∼98% test accuracy with near-perfect ROC–AUC.
- Engineered an end-to-end inference pipeline with facial landmark extraction, probability fusion, and time-based aggregation to distinguish sustained closure from blinking; deployed with OpenCV for live and offline monitoring.

Skills

Data Science & Analytics: Exploratory Data Analysis (EDA), Statistical Analysis, Hypothesis Testing, A/B Testing, Experimentation, Causal Inference, KPI Definition, Pandas, NumPy
Programming & Databases: Python, SQL, R, C++, Snowflake, PostgreSQL, ChromaDB
Machine Learning & Deep Learning: Scikit-learn, XGBoost, PyTorch, OpenCV, Hugging Face Transformers, CNNs, LSTMs, VGG16, BERT, GPT, Regression, Classification, Clustering, Time-Series Forecasting
Visualization, Cloud & LLM Systems: Power BI, Tableau, Streamlit, Matplotlib, Seaborn, Folium, R Shiny; AWS (S3, EC2), Docker; LLMs, RAG, Prompt Engineering, Multi-agent Systems